MSc Project Proposal

Interactive Web-Based Exploration of Algorithmic Pattern Generation Systems *(working title)*

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1. Project Aim

The aim of this project is to design and develop an intuitive web-based application that allows users to create and explore a wide variety of visual patterns through interactive algorithmic tools.

The system will enable users to experiment with different pattern-generation approaches in real time by manipulating rules, parameters, and generative systems. The interface will prioritise clarity and usability, allowing users to observe how patterns emerge step-by-step from underlying algorithms.

While the primary goal is creative exploration, the system will also support an educational dimension by helping users understand how different algorithmic structures influence visual pattern formation.

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2. Background and Motivation

This project builds upon my undergraduate research project:

"Maths to Magic and Visual Wizardry: Exploring the Procedural Creation of Islamic Geometrical Patterns in Houdini."

In that work, Islamic geometric patterns were generated using **shape grammar rules** within a procedural node-based modelling environment. The research explored how complex geometric patterns can emerge from relatively simple rule-based transformations.

However, the system was limited to a specialist modelling environment and focused primarily on a specific class of geometric patterns.

This MSc project aims to extend that work in several ways:

- Moving from procedural modelling software to a **fully coded browser-based system**
- Expanding beyond a single pattern tradition to support **multiple generative pattern systems**
- Creating an **interactive interface** that allows users to experiment directly with pattern rules and algorithms

The project is motivated by the idea that many visual patterns can be understood as the result of **algorithmic processes**, ranging from highly ordered geometric systems to more random or procedural pattern generation methods (noise type). By including multiple generative approaches, the project investigates how algorithmic rules influence visual pattern formation.

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3. Project Objectives

The main objectives of the project are:

- Design and implement a **web-based interactive pattern generation tool**
- Develop an intuitive interface for experimenting with **algorithmic pattern systems**
- Extend the principles of **shape grammar rules** explored in previous work
- Support multiple **generative pattern approaches**, including rule-based and procedural methods
- Provide **real-time visual feedback** that allows users to observe patterns forming step-by-step
- Allow users to customise parameters and manipulate pattern rules to create unique designs
- Investigate what constitutes a distinct **pattern type** within generative systems
- Explore how different algorithmic approaches influence the visual characteristics of patterns

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4. Research Questions

The project investigates how algorithmic systems can support the creation and exploration of patterns through interactive tools. The research will be guided by the following questions:

1. **How can rule-based systems such as shape grammars be structured to support the generation of a wide variety of geometric patterns?**

2. **What defines a distinct "pattern type" within generative pattern systems?**

3. **How do different algorithmic approaches (e.g., rule-based, recursive, or stochastic systems) influence pattern formation?**

4. **How can an interactive web interface help users understand the relationship between algorithmic rules and visual patterns?**

5. **How usable and engaging is the system for users of varying experience?**

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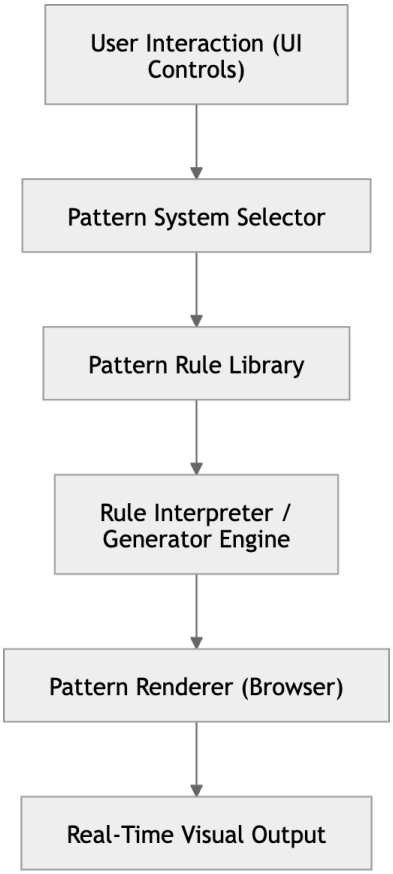
5. Technical Approach

The application will be developed as a browser-based system using **JavaScript and modern web graphics technologies** to support real-time interaction and rendering.

Key points:

- Real-time feedback for user interactions
- **Software design patterns** and **metaprogramming techniques** for modular rule architecture (*"code that writes code"*)
- Support for **multiple generative pattern systems** with an extensible framework

Figure 1: *System architecture showing modular flow from user input to real-time pattern rendering.*



5.1 System Architecture

The application will be structured as a **modular pattern generation pipeline** with several core components, designed to support multiple generative pattern systems and real-time interactive feedback:

User Interaction (UI Controls)
The user interacts with the system through intuitive controls, selecting pattern types, adjusting parameters, and exploring rule-based transformations.

Pattern System Selector
This component allows users to choose between different generative approaches, such as geometric (shape grammars), recursive (L-systems), or stochastic/procedural (noise-based) systems.

Pattern Rule Library
A collection of algorithmic rules and pattern definitions that describe how each pattern type is constructed. The library supports multiple rule sets, enabling a wide variety of pattern families.

Rule Interpreter / Generation Engine
This central engine interprets the stored rules and generates pattern geometry or procedural outputs. It translates algorithmic specifications into visual patterns in real time.

Pattern Renderer
Responsible for visualising the generated patterns within the browser, the renderer ensures that changes to parameters or rules are immediately reflected in the output.

Real-Time Visual Output
The system delivers live feedback to the user, allowing them to observe how rule modifications and parameter adjustments affect the resulting pattern step by step.

By separating these components, the architecture remains **modular and extensible**, allowing new pattern systems, rules, or rendering techniques to be added without restructuring the overall pipeline.

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6. Generative Pattern Systems

The project will support multiple pattern generation approaches along a continuum from **highly ordered geometric systems** to **stochastic/procedural systems**:

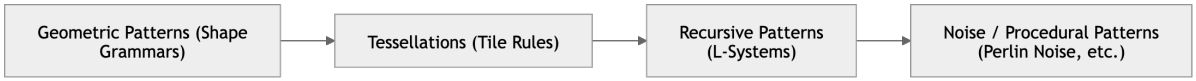
- **Geometric rule systems** (shape grammars, symmetry rules)
- **Recursive systems** (L-systems)
- **Procedural/noise-based systems** (Perlin noise, cellular automata)

Supporting multiple systems allows exploration of **how algorithmic logic shapes visual patterns** and strengthens the research contribution.

These different systems represent varying approaches to mapping mathematical rules to visual structure. Conceptually, they can be viewed as existing along a continuum between **order and randomness**, ranging from highly structured geometric tilings to more organic or stochastic pattern forms.

Supporting multiple generative approaches will allow the application to explore how different algorithmic structures influence pattern formation.

Figure 2: *Pattern Spectrum — continuum from Pure Order (Geometric Patterns) to Pure Randomness (Noise/Procedural Patterns)*



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7. Interface Design

A key component of the project is the design of a clear and intuitive interface.

The interface will aim to:

- Present pattern tools and parameters in a **visually understandable layout**
- Allow users to experiment with pattern rules without requiring programming knowledge
- Provide real-time feedback that illustrates how patterns evolve from algorithmic processes
- Encourage exploration and creative experimentation

Although the system is not primarily designed as an educational tool, the interface should naturally support learning by helping users understand the relationship between algorithms and visual outcomes.

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8. Evaluation

The project will be evaluated through a **user study** focusing on several aspects of the application.

Evaluation criteria:

- **Usability**: ease of use and clarity
- **Interface design**: effectiveness of layout and feedback
- **Functionality**: ability to generate intended pattern types
- **Performance**: responsiveness and real-time operation
- **Creative capability**: variety of patterns, user customisation
- **Educational value**: understanding rules-to-patterns relationships
- **User engagement**: enjoyment and exploration

User testing will also explore **what constitutes a distinct pattern type** and how effectively the system supports multiple pattern families. As algorithmic pattern systems can combine rules and parameters in many ways, the application has the potential to generate a very large number of possible pattern variations. The evaluation will therefore examine how effectively the system supports diverse pattern families and how scalable the architecture is for supporting new pattern types.

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9. Challenges and Risks

Several challenges may arise during development.

Defining Pattern Types
Generative systems can produce many visual variations, making it difficult to clearly define categories of pattern types.

Flexible Rule Architecture
Designing a system capable of supporting multiple pattern generation approaches while remaining maintainable.

Real-Time Performance
Ensuring the system remains responsive while generating complex patterns in the browser.

Balancing Simplicity and Flexibility
Designing an interface that remains intuitive while supporting advanced generative functionality.

These challenges will be addressed through iterative development and testing throughout the project.

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10. Expected Outcomes

The project will produce:

- A **functional web-based pattern generation application**
- A modular architecture supporting multiple generative pattern systems
- An intuitive interface for exploring algorithmic pattern creation
- Documentation describing the algorithms and system design
- Results from user testing, evaluating usability and creative potential

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References

Initial list of academic references, including some core from previous research:

1. Stiny, G., & Gips, J. (1972). *Shape Grammars and the Generative Specification of Painting and Sculpture*. Environment and Planning B, 1(2), 103–126.
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